

Networking Fundamentals

Network types, topologies, and the OSI model

Objectives 1.1 · 1.2 · 1.6

01 Core concepts

A **network** is two or more systems joined by a transmission medium and a shared set of protocols. Everything on it is either a **node** (a device that sends, receives, or forwards data) or a **link** (the path between nodes).

<i>Intermediate node</i>	Forwards traffic. Switches, routers.
<i>End system (host)</i>	Sends and receives traffic. PCs, phones, servers.
<i>Server / client</i>	A server provides resources; a client consumes them.

CLIENT-SERVER VS. PEER-TO-PEER

CENTRALIZED

Client-server

- Services managed and secured from one place
- Standard for businesses and enterprises
- Easier to control, back up, and scale
- Costs more; servers become critical points of failure

DECENTRALIZED

Peer-to-peer

- Every host is both client and server
- Common at home; a small one is a *workgroup*
- Cheap and simple, no dedicated server
- Hard to secure and manage as it grows

WATCH FOR

Hosts on a client-server network often play both roles, such as a web server acting as a client of a database. What makes it client-server is *central administration*, not what each machine does.

APPLIANCES, APPLICATIONS, FUNCTIONS

<i>Appliance</i>	A device built for one network job: forwarding (switch, router, access point), security (firewall, IDS), or performance (load balancer, proxy). Runs on its own hardware (<i>physical</i>) or as a VM on a hypervisor (<i>virtual</i>).
<i>Application</i>	The shared services the network exists to deliver, like file sharing and email.
<i>Function</i>	A capability layered on top. A VPN lets a device join the LAN securely over the Internet; QoS prioritizes time-sensitive traffic such as voice and video.

02 Network types

A network's type is defined by its **size** (number of nodes) and **scope** (the area it covers).

TYPE

DEFINING TRAITS

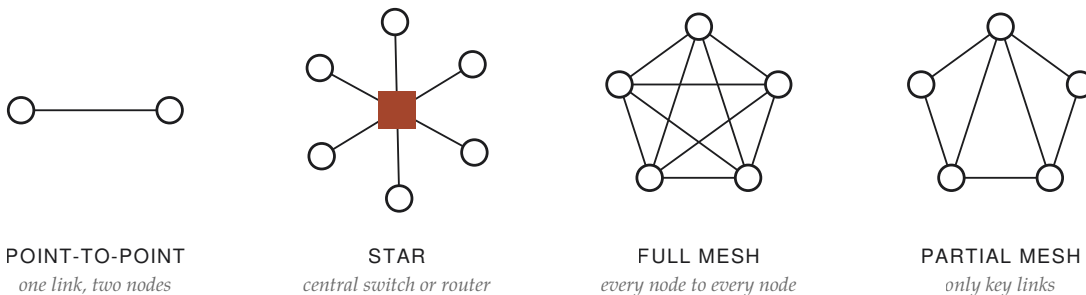
<i>LAN</i>	One location, directly connected by cable or short-range wireless, owned by one organization.
<i>Home / SOHO</i>	One all-in-one router/switch/AP. A SOHO may add a server.
<i>SME</i>	Dozens of users, structured cabling, multiple switches and routers.
<i>Enterprise</i>	Hundreds to thousands of hosts on enterprise-class gear.
<i>Datacenter</i>	Servers and storage only. No end-user devices.

WAN

A network of networks over long distances: branch offices, remote workers, datacenter links. Typically leased from a service provider.

03 Topologies

The **physical topology** is how devices are cabled. The **logical topology** is how data actually flows. Two networks can be wired differently yet behave identically.



TOPOLOGY

WHAT TO REMEMBER

<i>Point-to-point</i>	Exactly two nodes with guaranteed bandwidth. Can be logical: two WAN routers separated by many hops still only address each other. The 1:1 relationship is what defines it.
<i>Star</i>	The most common physical topology . Every node connects to a central device, which makes it easy to monitor, reconfigure, and troubleshoot. A fault is isolated to the cable, the NIC, or the central device.
<i>Hub-and-spoke</i>	Same shape as a star, but the term is used for WANs : headquarters as the hub, remote sites as spokes.
<i>Full mesh</i>	Every node linked to every other. Maximum redundancy, but the link count explodes.
<i>Partial mesh</i>	Only critical devices are interconnected. This is how real WANs and the Internet are built.

$$links = n(n - 1) / 2$$

4 nodes → 6 links · 40 nodes → 780 links · expect to calculate this

04 The OSI model

A seven-layer reference model from ISO. It's a **guideline** for designing and troubleshooting networks, not a standard that protocols must follow.

A **protocol** is a set of rules for exchanging data. Every protocol does two things: **addressing** (where data goes) and **encapsulation** (how it's packaged).

<i>Same-layer</i>	A layer communicates with the matching layer on the other node. Both must run the same protocol.
<i>Adjacent-layer</i>	On one node, each layer serves the layer above it and relies on the layer below.

THE SEVEN LAYERS

	LAYER	ITS JOB	PDU	EXAMPLES
7	<i>Application</i>	Where applications meet the network.	Data	HTTP, DNS, SMTP, FTP
6	<i>Presentation</i>	Translates data format: encoding, encryption, compression.	Data	TLS, JPEG, ASCII
5	<i>Session</i>	Opens, maintains, and closes conversations between hosts.	Data	RPC, NetBIOS
4	<i>Transport</i>	End-to-end delivery. Ports identify the application. TCP is reliable; UDP is fast.	Segment (TCP) Datagram (UDP)	TCP, UDP
3	<i>Network</i>	IP addressing and routing between networks.	Packet	IP, ICMP · router
2	<i>Data Link</i>	MAC addressing and delivery within the local segment.	Frame	Ethernet, Wi-Fi · switch, AP
1	<i>Physical</i>	Moves raw bits as signals across the media.	Bits	Cables · hub, repeater

MNEMONIC

1 → 7: *Please Do Not Throw Sausage Pizza Away*
 7 → 1: *All People Seem To Need Data Processing*

FAST RECALL

MAC is layer 2, IP is layer 3, ports are layer 4.
 Hub is layer 1, switch is layer 2, router is layer 3.

ENCAPSULATION

Sending data moves **down** the stack, and each layer wraps what it receives in its own header, like an envelope inside an envelope. Receiving moves **up** the stack and removes each header in turn; that's **decapsulation**. A web request looks like this:

HTTP data <i>Data</i>	+ TCP header <i>Segment</i>	+ IP header <i>Packet</i>	+ Ethernet header <i>Frame</i>	Signal on the wire <i>Bits</i>
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05 Check yourself

Main advantage of client-server over peer-to-peer?

Centralized management and security.

Physical vs. logical topology?

How it's wired vs. how data flows.

Most common physical topology?

Star.

A star layout used across a WAN is called?

Hub-and-spoke.

How many links in a full mesh of 6 nodes?

$6 \times 5 / 2 = 15$.

Which layers use MAC, IP, and port numbers?

2, 3, and 4.

PDU at the Transport layer?

Segment for TCP, datagram for UDP.

Which layer handles encryption and formatting?

Presentation, layer 6.